

D:\Aplikasi Kalkulus\calculus_simulator.html

```

1  <!DOCTYPE html>
2  <html lang="id">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>CALC_SIMS: Calculus Simulator</title>
7
8      <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
9      <script src="https://cdnjs.cloudflare.com/ajax/libs/mathjs/11.8.0/math.js"></script>
10     <!-- Tambahan Library CAS untuk Integral Analitik Eksak -->
11     <script src="https://cdn.jsdelivr.net/npm/nerdamer@1.1.13/nerdamer.core.js"></script>
12     <script src="https://cdn.jsdelivr.net/npm/nerdamer@1.1.13/Calculus.js"></script>
13
14     <link rel="preconnect" href="https://fonts.googleapis.com">
15     <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
16     <link href="https://fonts.googleapis.com/css2?
family=Montserrat:ital,wght@0,400;0,700;0,900;1,400;1,700;1,900&display=swap"
rel="stylesheet">
17
18     <style>
19         /* CSS Variables: P3 Reload Palette */
20         :root {
21             --p3-deep-blue: #02112e;
22             --p3-mid-blue: #0033cc;
23             --p3-cyan: #00e5ff;
24             --p3-white: #ffffff;
25             --p3-black: #000000;
26             --p3-pink: #cc0044;
27             --p3-gray: #1a1a1a;
28
29             --hard-shadow: 8px 8px 0px var(--p3-black);
30             --hard-shadow-cyan: 5px 5px 0px var(--p3-cyan);
31             --hard-shadow-pink: 5px 5px 0px var(--p3-pink);
32         }
33
34         /* Reset & Base Styles */
35         * {
36             box-sizing: border-box;
37             margin: 0;
38             padding: 0;
39             font-family: 'Montserrat', sans-serif;
40         }
41
42         /* =====
43         GLOBAL SCROLLBAR (Persona 3 Style)
44         ===== */
45         html {
46             scrollbar-width: thin;

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47     scrollbar-color: var(--p3-cyan) var(--p3-deep-blue);
48 }
49
50 ::-webkit-scrollbar { width: 16px; }
51 ::-webkit-scrollbar-track {
52     background: var(--p3-deep-blue);
53     border-left: 4px solid var(--p3-black);
54 }
55 ::-webkit-scrollbar-thumb {
56     background: var(--p3-cyan);
57     border: 4px solid var(--p3-black);
58     border-right: none;
59     cursor: pointer;
60 }
61 ::-webkit-scrollbar-thumb:hover { background: var(--p3-pink); }
62
63 body {
64     background: linear-gradient(180deg, var(--p3-deep-blue) 0%, #001b5e 100%);
65     color: var(--p3-black);
66     line-height: 1.6;
67     min-height: 100vh;
68     overflow-x: hidden;
69     position: relative;
70     z-index: 0;
71     padding: 40px 20px;
72     background-attachment: fixed;
73 }
74
75 /* =====
76 P3 BACKGROUND EFFECTS (Ocean & God Rays)
77 ===== */
78
79 /* Lapisan 1: Pendaran lautan dalam (Caustics Glow) */
80 body::after {
81     content: "";
82     position: fixed;
83     top: 0; left: 0; right: 0; bottom: 0;
84     background:
85         radial-gradient(circle at 30% 50%, rgba(0, 229, 255, 0.08) 0%, transparent
50%),
86         radial-gradient(circle at 70% 80%, rgba(204, 0, 68, 0.04) 0%, transparent
50%);
87     animation: oceanBreathe 8s ease-in-out infinite alternate;
88     z-index: -4;
89     pointer-events: none;
90 }
91
92 @keyframes oceanBreathe {
93     0% { opacity: 0.5; transform: scale(1); }
94     100% { opacity: 1; transform: scale(1.1); }

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95     }
96
97     /* Lapisan 2: Titik-titik Halftone */
98     body::before {
99         content: "";
100        position: fixed;
101        top: 0; left: 0; right: 0; bottom: 0;
102        background-image: radial-gradient(rgba(0, 229, 255, 0.15) 1px, transparent 1px);
103        background-size: 15px 15px;
104        z-index: -3;
105        animation: moveDots 2.5s linear infinite;
106    }
107
108    @keyframes moveDots {
109        from { background-position: 0px 0px; }
110        to { background-position: 15px 15px; }
111    }
112
113    /* Lapisan 3: God Rays */
114    .god-rays {
115        position: fixed;
116        top: 0; left: 0; width: 150vw; height: 150vh;
117        transform-origin: 0% 0%;
118        background:
119            radial-gradient(circle at 0% 0%, rgba(255, 255, 255, 0.15) 0%, rgba(0, 229,
120            255, 0.05) 20%, transparent 50%),
121            conic-gradient(
122                from 90deg at 0% 0%,
123                transparent 0deg, rgba(255, 255, 255, 0.05) 15deg,
124                transparent 35deg, rgba(0, 229, 255, 0.08) 50deg,
125                transparent 70deg, rgba(255, 255, 255, 0.03) 85deg,
126                transparent 100deg
127            );
128        -webkit-mask-image: radial-gradient(circle at 0% 0%, black 0%, black 20%,
129        transparent 70%);
130        mask-image: radial-gradient(circle at 0% 0%, black 0%, black 20%, transparent
131        70%);
132        animation: swayRays 12s ease-in-out infinite alternate;
133        z-index: -2;
134        pointer-events: none;
135        mix-blend-mode: screen;
136    }
137
138    @keyframes swayRays {
139        0% { transform: rotate(-1deg); opacity: 0.3; }
140        100% { transform: rotate(3deg); opacity: 0.6; }
141    }
142
143    /* Lapisan 4: Teks Raksasa Latar Belakang */
144    .bg-text {
145        position: fixed; font-size: 22vw; font-weight: 900;

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143     font-style: italic; color: transparent;
144     -webkit-text-stroke: 2px rgba(0, 229, 255, 0.1);
145     z-index: -1; pointer-events: none;
146     line-height: 0.8; text-transform: uppercase;
147 }
148 .bg-text-1 { top: 15%; left: -5%; transform: skewX(-15deg) rotate(-5deg); }
149 .bg-text-2 { bottom: 20%; right: -5%; transform: skewX(-15deg) rotate(5deg); }
150
151 h1, h2, h3 { font-weight: 900; text-transform: uppercase; font-style: italic; }
152
153 /* =====
154     MAIN MENU STYLES
155     ===== */
156 #menu-container {
157     display: flex; flex-direction: column; align-items: center; justify-content:
center;
158     min-height: 80vh; position: relative; z-index: 10;
159 }
160
161 .menu-title-block { text-align: center; margin-bottom: 60px; position: relative;
transform: skewX(-10deg); }
162 .menu-title-block h1 {
163     font-size: 6rem; color: var(--p3-white);
164     text-shadow: 8px 8px 0px var(--p3-black), -3px -3px 0px var(--p3-pink);
165     letter-spacing: -3px; line-height: 1;
166 }
167 .menu-title-block h2 {
168     font-size: 1.5rem; color: var(--p3-black); background-color: var(--p3-cyan);
169     display: inline-block; padding: 8px 30px; margin-top: 10px;
170     box-shadow: 5px 5px 0px var(--p3-black); border: 4px solid var(--p3-black);
letter-spacing: 2px;
171 }
172
173 .menu-btn-group { display: flex; flex-direction: column; gap: 20px; align-items:
center; position: relative; }
174 .menu-btn {
175     background-color: var(--p3-black); color: var(--p3-white);
176     border: 4px solid var(--p3-black); padding: 15px 50px; font-size: 2rem;
177     font-weight: 900; font-style: italic; text-transform: uppercase; cursor:
pointer;
178     transition: all 0.2s cubic-bezier(0.175, 0.885, 0.32, 1.275);
179     box-shadow: 6px 6px 0px var(--p3-cyan); transform: skewX(-15deg);
180     width: 350px; text-align: left; position: relative; overflow: hidden;
181 }
182 .menu-btn::before {
183     content: ''; position: absolute; top: 0; left: -100%;
184     width: 100%; height: 100%; background-color: var(--p3-cyan);
185     transition: left 0.3s ease; z-index: 1;
186 }
187 .menu-btn span { position: relative; z-index: 2; }
188 .menu-btn:hover {

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189     color: var(--p3-black); transform: skewX(-15deg) scale(1.05) translateX(10px);
190     box-shadow: 8px 8px 0px var(--p3-pink);
191 }
192 .menu-btn:hover::before { left: 0; }
193
194 /* Modals (Pop-up) */
195 .modal-overlay {
196     position: fixed; top: 0; left: 0; width: 100%; height: 100%;
197     background: rgba(2, 17, 46, 0.9); backdrop-filter: blur(5px);
198     z-index: 9999; display: flex; align-items: center; justify-content: center;
199     opacity: 0; pointer-events: none; transition: opacity 0.3s ease;
200 }
201 .modal-overlay.active { opacity: 1; pointer-events: auto; }
202 .modal-content {
203     background-color: var(--p3-white); border: 5px solid var(--p3-black);
204     padding: 40px 30px; max-width: 600px; width: 90%;
205     transform: skewX(-5deg) translateY(50px); box-shadow: 15px 15px 0px var(--p3-
cyan);
206     position: relative; transition: transform 0.3s cubic-bezier(0.175, 0.885, 0.32,
1.275);
207 }
208 .modal-overlay.active .modal-content { transform: skewX(-5deg) translateY(0); }
209 .modal-header {
210     background-color: var(--p3-black); color: var(--p3-cyan); display: inline-block;
211     padding: 10px 25px; font-size: 2rem; margin-top: -65px; margin-bottom: 20px;
212     margin-left: -20px; box-shadow: 5px 5px 0px var(--p3-pink); border: 3px solid
var(--p3-black);
213 }
214 .modal-body {
215     font-size: 1.1rem; font-weight: 700; color: var(--p3-black); line-height: 1.8;
216     max-height: 400px; overflow-y: auto; padding-right: 15px;
217 }
218 .modal-body ul { margin-left: 20px; margin-top: 10px; }
219 .modal-body li { margin-bottom: 10px; }
220 .code-snippet {
221     background-color: var(--p3-gray); color: var(--p3-cyan);
222     padding: 2px 8px; font-family: monospace; font-style: normal; border: 2px solid
var(--p3-black);
223 }
224 .dev-list { list-style: none; margin: 0; padding: 0; }
225 .dev-item {
226     background: #f4f4f4; border: 3px solid var(--p3-black); padding: 15px; margin-
bottom: 15px;
227     border-left: 10px solid var(--p3-pink);
228 }
229 .dev-item h4 { font-size: 1.3rem; margin-bottom: 5px; color: var(--p3-mid-blue); }
230 .dev-item p { font-size: 1rem; font-weight: 800; }
231 .btn-close {
232     position: absolute; top: -20px; right: -20px; background-color: var(--p3-pink);
233     color: var(--p3-white); border: 4px solid var(--p3-black); width: 50px; height:
50px;

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234         font-size: 1.5rem; font-weight: 900; cursor: pointer; box-shadow: 4px 4px 0px
var(--p3-black); transition: transform 0.1s;
235     }
236     .btn-close:hover { transform: scale(1.1); background-color: var(--p3-white); color:
var(--p3-pink); }
237
238     /* =====
239     APP STYLES
240     ===== */
241     .app-header { max-width: 1300px; margin: 0 auto 50px auto; position: relative; }
242     .header-title-container { position: relative; display: inline-block; transform:
skewX(-15deg); }
243     .header-bg-block {
244         position: absolute; top: 10px; left: -15px; width: 100%; height: 100%;
245         background-color: var(--p3-cyan); z-index: 1;
246     }
247     .app-header h1 {
248         position: relative; font-size: 4.5rem; color: var(--p3-white);
249         text-shadow: 6px 6px 0px var(--p3-black); z-index: 2; letter-spacing: -2px;
padding: 0 20px;
250     }
251     .app-header p {
252         color: var(--p3-black); background-color: var(--p3-white); font-size: 1.2rem;
253         font-weight: 900; display: inline-block; padding: 5px 20px;
254         transform: skewX(-15deg) translateY(-10px); margin-left: 40px;
255         box-shadow: 4px 4px 0px var(--p3-cyan); border: 3px solid var(--p3-black);
256     }
257
258     .main-container {
259         max-width: 1300px; margin: 0 auto; display: flex; gap: 40px;
260         align-items: stretch; padding-bottom: 100px;
261     }
262     .left-column { flex: 0 0 35%; display: flex; flex-direction: column; gap: 40px; }
263     .right-column { flex: 1; display: flex; flex-direction: column; gap: 40px; }
264
265     .card {
266         background-color: var(--p3-white); padding: 30px; position: relative;
267         border: 4px solid var(--p3-black); box-shadow: var(--hard-shadow);
268     }
269     .card::before {
270         content: ''; position: absolute; top: -4px; right: -4px;
271         border-top: 30px solid var(--p3-deep-blue); border-left: 30px solid transparent;
z-index: 5;
272     }
273     .card::after {
274         content: ''; position: absolute; top: -4px; right: -4px;
275         border-top: 36px solid var(--p3-cyan); border-left: 36px solid transparent; z-
index: 4;
276     }
277
278     .card-header { margin-bottom: 30px; display: flex; align-items: center; }

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```
279 .card-header h2, .card-header h3 {
280     font-size: 1.8rem; color: var(--p3-white); background-color: var(--p3-black);
281     padding: 8px 25px; transform: skewX(-15deg) translateX(-45px);
282     box-shadow: var(--hard-shadow-cyan); position: relative; z-index: 10;
283 }
284 .card-header h3 { box-shadow: var(--hard-shadow-pink); }
285
286 .tabs { display: flex; flex-direction: column; gap: 15px; margin-bottom: 40px;
position: relative; }
287 .tabs::before {
288     content: ''; position: absolute; left: 15px; top: 0; bottom: 0;
289     width: 4px; background-color: var(--p3-black); z-index: 1;
290 }
291 .tab-btn {
292     align-self: flex-start; padding: 10px 20px 10px 40px; border: 3px solid var(--
p3-black);
293     background: var(--p3-white); font-size: 1.1rem; font-weight: 900; font-style:
italic;
294     color: var(--p3-black); text-transform: uppercase; cursor: pointer; transform:
skewX(-15deg);
295     transition: all 0.2s cubic-bezier(0.175, 0.885, 0.32, 1.275);
296     box-shadow: 4px 4px 0px var(--p3-black); position: relative; z-index: 2;
297 }
298 .tab-btn:nth-child(2) { margin-left: 20px; }
299 .tab-btn:hover { transform: skewX(-15deg) translateX(10px); background-color: var(--
p3-black); color: var(--p3-cyan); }
300 .tab-btn.active {
301     background-color: var(--p3-cyan); color: var(--p3-black); border-color: var(--
p3-black);
302     box-shadow: inset 0px 0px 0px 3px var(--p3-white), 6px 6px 0px var(--p3-black);
padding-left: 50px;
303 }
304
305 .formula-info-box {
306     background-color: var(--p3-gray); border: 3px solid var(--p3-black);
307     padding: 15px; margin-bottom: 25px; transform: skewX(-5deg);
308     box-shadow: 4px 4px 0px var(--p3-black); color: var(--p3-white);
309 }
310 .formula-info-box .badge {
311     background-color: var(--p3-pink); color: var(--p3-white); font-size: 0.7rem;
312     font-weight: 900; padding: 2px 8px; margin-right: 10px; display: inline-block;
313     border: 2px solid var(--p3-black); transform: skewX(-10deg);
314 }
315 .formula-info-box .badge.cyan { background-color: var(--p3-cyan); color: var(--p3-
black); }
316 .formula-info-box i { font-family: 'Times New Roman', Times, serif; font-size:
1.1rem; letter-spacing: 1px; color: var(--p3-cyan); }
317 .formula-line { margin-bottom: 8px; display: flex; align-items: center;}
318 .formula-line:last-child { margin-bottom: 0; }
319
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```
320     .form-group { margin-bottom: 25px; display: flex; flex-direction: column; position:
relative; }
321     .form-group label {
322         font-size: 0.9rem; font-weight: 900; font-style: italic; color: var(--p3-white);
323         background-color: var(--p3-mid-blue); padding: 6px 15px; display: inline-block;
324         align-self: flex-start; transform: skewX(-15deg) translateY(10px)
translateX(10px);
325         z-index: 2; border: 2px solid var(--p3-black);
326     }
327     .form-group input {
328         padding: 18px 15px 12px 25px; border: 3px solid var(--p3-black); font-size:
1.2rem;
329         font-weight: 700; background-color: #f4f4f4; transform: skewX(-5deg);
transition: all 0.2s ease; color: var(--p3-black);
330     }
331     .form-group input:focus {
332         outline: none; background-color: var(--p3-white); border-color: var(--p3-cyan);
333         box-shadow: 6px 6px 0px var(--p3-black); transform: skewX(-5deg)
translateY(-2px);
334     }
335
336     .action-area { margin-top: 50px; }
337     .btn-primary {
338         width: 100%; background-color: var(--p3-black); color: var(--p3-white);
339         border: 4px solid var(--p3-cyan); padding: 20px 24px; font-size: 1.8rem;
340         font-weight: 900; font-style: italic; text-transform: uppercase; cursor:
pointer;
341         transition: all 0.1s ease; box-shadow: 8px 8px 0px var(--p3-cyan);
342         transform: skewX(-10deg); letter-spacing: 2px; position: relative; overflow:
hidden;
343     }
344     .btn-primary::before { content: '>>'; position: absolute; left: 20px; color: var(--
p3-cyan); transition: left 0.2s; }
345     .btn-primary:hover {
346         background-color: var(--p3-cyan); color: var(--p3-black); border-color: var(--
p3-black);
347         box-shadow: 10px 10px 0px var(--p3-black); transform: skewX(-10deg)
translate(-2px, -2px);
348     }
349     .btn-primary:hover::before { left: 30px; color: var(--p3-black); }
350     .btn-primary:active { transform: skewX(-10deg) translate(4px, 4px); box-shadow: 2px
2px 0px var(--p3-black); }
351
352     .error-message {
353         color: var(--p3-white); background-color: var(--p3-pink); padding: 12px;
354         font-weight: 900; font-style: italic; margin-top: 20px; text-align: center;
355         border: 3px solid var(--p3-black); transform: skewX(-10deg); display: none; box-
shadow: 4px 4px 0px var(--p3-black);
356     }
357
358     .result-summary {
```

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359     background-color: var(--p3-black); color: var(--p3-white); padding: 25px;
360     margin-bottom: 25px; border: 3px solid var(--p3-cyan); transform: skewX(-5deg);
361 }
362 .result-item {
363     font-size: 1.1rem; font-weight: 800; font-style: italic; display: flex;
364     justify-content: space-between; align-items: center; padding: 10px 0;
365     border-bottom: 2px dotted rgba(0, 229, 255, 0.3);
366 }
367 .result-item:last-child { border-bottom: none; }
368 .result-item span { color: var(--p3-cyan); font-size: 1.8rem; font-weight: 900;
text-shadow: 2px 2px 0px rgba(0,0,0,0.5); }
369
370 .math-container {
371     background-color: var(--p3-gray); border: 3px solid var(--p3-black);
372     padding: 25px; min-height: 120px; max-height: 400px; color: var(--p3-cyan);
373     font-weight: 600; font-size: 1rem; line-height: 1.8; text-align: left;
374     overflow-y: auto; overflow-x: hidden; position: relative;
375 }
376 .math-container::before {
377     content: 'SYSTEM.CALC_LOG'; display: block; font-size: 0.7rem;
378     color: var(--p3-white); margin-bottom: 10px; opacity: 0.5; letter-spacing: 2px;
379 }
380 .math-container i { color: var(--p3-white); font-family: 'Times New Roman', Times,
serif; font-size: 1.15rem; margin-left: 10px; }
381 .math-container strong { color: var(--p3-pink); display: block; margin-top: 20px; }
382
383 .charts-container { display: grid; grid-template-columns: 1fr 1fr; gap: 25px; }
384 .chart-wrapper {
385     background-color: var(--p3-white); padding: 15px; border: 4px solid var(--p3-
black);
386     position: relative; height: 250px; width: 100%; box-shadow: 6px 6px 0px
rgba(0,0,0,0.1);
387 }
388 .chart-label {
389     position: absolute; top: -15px; left: 10px; background: var(--p3-black);
390     color: var(--p3-cyan); border: 2px solid var(--p3-cyan); padding: 4px 15px;
391     font-size: 0.9rem; font-weight: 900; font-style: italic; transform:
skewX(-15deg); z-index: 10;
392 }
393
394 /* =====
395 FLOATING WIDGET (Live Simulation Pop-up)
396 ===== */
397 .floating-widget {
398     position: fixed; bottom: 30px; right: 30px; width: 500px; min-width: 350px; min-
height: 150px;
399     background-color: var(--p3-white); border: 4px solid var(--p3-black);
400     box-shadow: 10px 10px 0px rgba(0,0,0,0.8), 0 0 20px rgba(0, 229, 255, 0.4);
401     z-index: 9000; transform: skewX(-2deg); display: flex; flex-direction: column;
overflow: hidden;
402 }

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403
404     .resizer { position: absolute; bottom: 0; width: 25px; height: 25px; background-
color: var(--p3-cyan); z-index: 10; }
405     .resizer-br { right: 0; cursor: se-resize; clip-path: polygon(100% 0, 100% 100%, 0
100%); }
406     .resizer-bl { left: 0; cursor: sw-resize; clip-path: polygon(0 0, 100% 100%, 0
100%); }
407     .floating-widget.minimized .resizer { display: none; }
408
409     .floating-header {
410         background-color: var(--p3-black); color: var(--p3-cyan); padding: 10px 15px;
411         cursor: grab; display: flex; justify-content: space-between; align-items:
center;
412         border-bottom: 4px solid var(--p3-black); box-shadow: inset 0px -3px 0px rgba(0,
229, 255, 0.3); user-select: none;
413     }
414     .floating-header:active { cursor: grabbing; }
415     .floating-header h3 { margin: 0; font-size: 1.2rem; text-shadow: 2px 2px 0px var(--
p3-pink); letter-spacing: 1px; transform: skewX(-5deg); }
416     .floating-controls { display: flex; gap: 10px; }
417     .floating-controls button {
418         background: var(--p3-pink); color: var(--p3-white); border: 2px solid var(--p3-
white);
419         width: 30px; height: 30px; font-weight: 900; font-size: 1.2rem; cursor: pointer;
420         box-shadow: 2px 2px 0px var(--p3-cyan); display: flex; align-items: center;
justify-content: center; line-height: 0; padding-bottom: 3px;
421     }
422     .floating-controls button:hover { background: var(--p3-cyan); color: var(--p3-
black); box-shadow: 2px 2px 0px var(--p3-pink); }
423
424     .floating-body { padding: 15px; background-color: var(--p3-white); overflow: hidden;
flex: 1; display: flex; flex-direction: column; }
425     .floating-widget.minimized .floating-body { display: none; }
426     .floating-widget.minimized { width: 300px !important; height: auto !important; }
427
428     .simulation-container { padding: 5px 0; flex: 1; display: flex; flex-direction:
column; }
429     .track {
430         background-color: var(--track-bg); flex: 1; min-height: 100px;
431         border: 4px solid var(--p3-black); position: relative; overflow: hidden;
432         display: flex; align-items: center; box-shadow: inset 0 10px 30px
rgba(0,0,0,0.8); transform: skewX(-5deg);
433     }
434     .track::before {
435         content: ''; position: absolute; inset: 0;
436         background-image: linear-gradient(rgba(0, 229, 255, 0.1) 1px, transparent 1px),
linear-gradient(90deg, rgba(0, 229, 255, 0.1) 1px, transparent 1px);
437         background-size: 20px 20px; z-index: 1;
438     }
439     .road-line {
440         position: absolute; top: 50%; left: 0; right: 0; height: 6px; transform:
translateY(-50%);

```

```
441         background-image: linear-gradient(to right, var(--p3-cyan) 50%, transparent
50%);
442         background-size: 80px 100%; background-repeat: repeat-x; z-index: 2; box-shadow:
0 0 10px var(--p3-cyan);
443     }
444     .car-emoji {
445         font-size: 4rem; position: absolute; left: 20px; top: 50%;
446         transform: translateY(-50%) skewX(5deg) scaleX(-1); z-index: 5; user-select:
none;
447         filter: drop-shadow(-6px 6px 0px rgba(0,0,0,0.9));
448     }
449
450     .dec-star {
451         position: absolute; width: 20px; height: 20px; background-color: var(--p3-pink);
452         clip-path: polygon(50% 0%, 61% 35%, 98% 35%, 68% 57%, 79% 91%, 50% 70%, 21% 91%,
32% 57%, 2% 35%, 39% 35%); z-index: 10;
453     }
454
455     .btn-return {
456         background-color: var(--p3-black); color: var(--p3-white); border: 3px solid
var(--p3-cyan);
457         padding: 10px 20px; font-size: 1.2rem; font-weight: 900; font-style: italic;
cursor: pointer;
458         position: absolute; top: 0; right: 0; transform: skewX(-15deg); box-shadow: 4px
4px 0px var(--p3-black); z-index: 50;
459     }
460     .btn-return:hover { background-color: var(--p3-pink); border-color: var(--p3-black);
}
461
462     @media (max-width: 1024px) {
463         .main-container { flex-direction: column; padding-bottom: 200px; }
464         .left-column, .right-column { width: 100%; }
465         .app-header h1 { font-size: 2.8rem; }
466         .menu-title-block h1 { font-size: 4rem; }
467         .menu-btn { width: 100%; font-size: 1.5rem; }
468         .charts-container { grid-template-columns: 1fr; }
469         .chart-wrapper { height: 300px; margin-top: 25px; }
470         .card-header h2, .card-header h3 { transform: skewX(-15deg) translateX(-20px); }
471         .floating-widget { width: 90%; right: 5%; bottom: 20px; }
472     }
473 </style>
474 </head>
475 <body>
476
477     <!-- God Rays Background -->
478     <div class="god-rays"></div>
479
480     <!-- P3 Background Typography -->
481     <div class="bg-text bg-text-1">CALCULUS</div>
482     <div class="bg-text bg-text-2">SIMULATOR</div>
483
```

```

484 <!-- ===== -->
485 <!-- MENU UTAMA (LANDING PAGE) -->
486 <!-- ===== -->
487 <div id="menu-container">
488   <div class="menu-title-block">
489     <h1>CALC_SIMS</h1>
490     <h2>CALCULUS SIMULATOR: PERHITUNGAN JARAK TEMPUH</h2>
491   </div>
492
493   <div class="menu-btn-group">
494     <button class="menu-btn" onclick="showApp()"><span>> MULAI</span></button>
495     <button class="menu-btn" onclick="showModal('modal-panduan')"><span>>
PANDUAN</span></button>
496     <button class="menu-btn" onclick="showModal('modal-developer')"><span>>
DEVELOPER</span></button>
497   </div>
498 </div>
499
500 <!-- ===== -->
501 <!-- POP-UP / MODALS -->
502 <!-- ===== -->
503
504 <!-- Modal Panduan -->
505 <div id="modal-panduan" class="modal-overlay">
506   <div class="modal-content">
507     <button class="btn-close" onclick="closeModal('modal-panduan')">X</button>
508     <h2 class="modal-header">PANDUAN INPUT</h2>
509     <div class="modal-body">
510       <p>Ikuti format pengisian berikut untuk mencegah <i>System Error</i>:</p>
511       <ul>
512         <li>Gunakan <strong>Titik (.)</strong> untuk angka desimal, bukan koma.
<br><i>Contoh: 2.5</i></li>
513         <li>Waktu Akhir ( $t_1$ ) <strong>wajib</strong> lebih besar dari Waktu Awal
( $t_0$ ).</li>
514       </ul>
515       <br>
516       <p><strong>Bagi Mode FUNGSI CUSTOM:</strong></p>
517       <p>Masukkan fungsi terhadap waktu <span class="code-snippet">t</span>.
Gunakan operator matematika standar:</p>
518       <ul>
519         <li>Perkalian: gunakan bintang <span class="code-snippet">*</span> <br>
<i>(contoh: <span class="code-snippet">2*t</span>)</i></li>
520         <li>Pembagian: gunakan garis miring <span class="code-snippet">/</span>
<br><i>(contoh: <span class="code-snippet">t/2</span>)</i></li>
521         <li>Pangkat: gunakan simbol topi <span class="code-snippet">^</span>
<br><i>(contoh: <span class="code-snippet">t^2</span> untuk t kuadrat)</i></li>
522         <li>Penjumlahan/Pengurangan: <span class="code-snippet">+</span> dan
<span class="code-snippet">-</span> <br><i>(contoh: <span class="code-snippet">3*t^2 - 2*t +
5</span>)</i></li>
523       </ul>
524     </div>

```

```
525     </div>
526 </div>
527
528 <!-- Modal Developer -->
529 <div id="modal-developer" class="modal-overlay">
530     <div class="modal-content">
531         <button class="btn-close" onclick="closeModal('modal-developer')">X</button>
532         <h2 class="modal-header">DEVELOPER</h2>
533         <div class="modal-body">
534             <ul class="dev-list">
535                 <li class="dev-item">
536                     <h4>1. Zumar Abid Zaidan</h4>
537                     <p>NIM: D1041251038</p>
538                 </li>
539                 <li class="dev-item">
540                     <h4>2. Muhammad Ridho Maulana</h4>
541                     <p>NIM: D1041251012</p>
542                 </li>
543                 <li class="dev-item">
544                     <h4>3. Khaira Shafa Arkomah</h4>
545                     <p>NIM: D1041251103</p>
546                 </li>
547             </ul>
548         </div>
549     </div>
550 </div>
551
552 <!-- ===== -->
553 <!-- APP CONTAINER -->
554 <!-- ===== -->
555 <div id="app-container" style="display: none;">
556
557     <!-- Header Aplikasi -->
558     <header class="app-header">
559         <div class="header-title-container">
560             <div class="header-bg-block"></div>
561             <h1>CALC_SIMS</h1>
562         </div>
563         <br>
564         <p>CALCULUS SIMULATOR: JARAK TEMPUH KENDARAAN</p>
565         <!-- Tombol Kembali ke Menu -->
566         <button class="btn-return" onclick="showMenu()">< KEMBALI</button>
567     </header>
568
569     <!-- Kontainer Utama -->
570     <main class="main-container">
571
572         <!-- KOLOM KIRI (Panel Input) -->
573         <aside class="left-column">
574             <div class="card">
```

```

575     <div class="dec-star" style="top: -10px; left: -10px;"></div>
576     <div class="card-header">
577         <h2>PARAMETERS</h2>
578     </div>
579
580     <!-- Mode Switcher -->
581     <div class="tabs" id="mode-switcher">
582         <button class="tab-btn active" id="tab-fisika"
onclick="setMode('fisika')">PHYSICS_MODE</button>
583         <button class="tab-btn" id="tab-custom"
onclick="setMode('custom')">CUSTOM_FUNC</button>
584     </div>
585
586     <!-- Formula Info Box -->
587     <div class="formula-info-box">
588         <div class="formula-line">
589             <span class="badge cyan">UMUM</span>
590             <i> $s = s_0 + \int v(t) dt$ </i>
591         </div>
592         <div id="physics-formula-ref" class="formula-line" style="margin-
top: 10px;">
593             <span class="badge">GLBB</span>
594             <i> $s = s_0 + v_0 t + \frac{1}{2} a t^2$ </i>
595         </div>
596         <div id="custom-formula-ref" class="formula-line" style="display:
none; margin-top: 10px; flex-direction: column; align-items: flex-start; gap: 5px;">
597             <div><span class="badge">CUSTOM</span> <i> $v(t) = at^2 + bt +
c$ </i></div>
598             <!-- MODIFIKASI: Mengubah Tampilan menjadi Integral Tentu -->
599             <div style="padding-left: 73px;"><i> $s = s_0 + \int_{t_0}^{t_1} v(t) dt$ </i></div>
600             </div>
601         </div>
602
603     <!-- Form Container -->
604     <div id="form-container" class="form-container">
605
606         <!-- Form Parameter Fisika -->
607         <form id="form-fisika">
608             <div class="form-group">
609                 <label for="input-v0">KECEPATAN AWAL ( $v_0$ ) [m/s]</label>
610                 <input type="number" id="input-v0" placeholder="0"
step="any" required>
611             </div>
612             <div class="form-group">
613                 <label for="input-a">PERCEPATAN ( $a$ ) [ $m/s^2$ ]</label>
614                 <input type="number" id="input-a" placeholder="2.5"
step="any" required>
615             </div>
616             <div class="form-group">
617                 <label for="input-t0">WAKTU AWAL ( $t_0$ ) [s]</label>

```

```

618         <input type="number" id="input-t0" placeholder="0"
step="any" required>
619     </div>
620     <div class="form-group">
621         <label for="input-t1">WAKTU AKHIR ( $t_1$ ) [s]</label>
622         <input type="number" id="input-t1" placeholder="10"
step="any" required>
623     </div>
624     <div class="form-group">
625         <label for="input-s0">POSISI AWAL ( $s_0$ ) [m]</label>
626         <input type="number" id="input-s0" placeholder="0"
step="any" required>
627     </div>
628 </form>
629
630 <!-- Form Fungsi Custom -->
631 <form id="form-custom" style="display: none;">
632     <div class="form-group">
633         <label for="input-v-func">FUNGSI KECEPATAN  $v(t)$  [m/s]
</label>
634         <input type="text" id="input-v-func" placeholder="Contoh:
5*t^2 + 2*t + 100" required>
635     </div>
636     <div class="form-group">
637         <label for="input-tc0">WAKTU AWAL ( $t_0$ ) [s]</label>
638         <input type="number" id="input-tc0" placeholder="0"
step="any" required>
639     </div>
640     <div class="form-group">
641         <label for="input-tc1">WAKTU AKHIR ( $t_1$ ) [s]</label>
642         <input type="number" id="input-tc1" placeholder="100"
step="any" required>
643     </div>
644     <div class="form-group">
645         <label for="input-sc0">POSISI AWAL ( $s_0$ ) [m]</label>
646         <input type="number" id="input-sc0" placeholder="0"
step="any" required>
647     </div>
648 </form>
649 </div>
650
651 <!-- Action Area -->
652 <div class="action-area">
653     <button id="btn-hitung" class="btn-primary"
onclick="mulaiSimulasi()">EXECUTE</button>
654     <!-- Teks Error Validasi -->
655     <p id="error-msg" class="error-message">ERROR: INPUT TIDAK VALID!
</p>
656 </div>
657 </div>
658 </aside>

```

```

659
660 <!-- KOLOM KANAN (Panel Output & Visualisasi) -->
661 <section class="right-column">
662
663 <!-- Card 1: Hasil Analisis Matematis -->
664 <div class="card">
665 <div class="card-header">
666 <h3>ANALYSIS_RESULT</h3>
667 </div>
668
669 <div class="result-summary">
670 <div class="result-item">
671 TOTAL JARAK
672 <div><span id="output-jarak">-</span> m</div>
673 </div>
674 <div class="result-item">
675 POSISI AKHIR
676 <div><span id="output-posisi">-</span> m</div>
677 </div>
678 </div>
679
680 <!-- Wadah Rumus/Langkah Penyelesaian -->
681 <div id="langkah-penyelesaian" class="math-container">
682 > MENUNGGU INPUT PARAMETER...
683 </div>
684 </div>
685
686 <!-- Card 2: Visualisasi Grafik -->
687 <div class="card">
688 <div class="dec-star" style="top: -10px; right: -10px; background-color:
var(--p3-cyan);"></div>
689 <div class="card-header">
690 <h3>DATA_VISUALIZATION</h3>
691 </div>
692
693 <div class="charts-container">
694 <div class="chart-wrapper">
695 <div class="chart-label">CHART:  $v(t)$ </div>
696 <canvas id="chart-velocity"></canvas>
697 </div>
698 <div class="chart-wrapper">
699 <div class="chart-label">CHART:  $s(t)$ </div>
700 <canvas id="chart-position"></canvas>
701 </div>
702 </div>
703 </div>
704 </section>
705 </main>
706
707 <!-- FLOATING WIDGET: LIVE SIMULATION -->

```

```

708 <div id="sim-widget" class="floating-widget" style="display: none;">
709   <div class="floating-header" id="sim-widget-header">
710     <h3>LIVE_SIMULATION</h3>
711     <div class="floating-controls">
712       <button id="btn-minimize-sim" onclick="toggleSimWidget()">-</button>
713     </div>
714   </div>
715   <div class="floating-body" id="sim-widget-body">
716     <div class="simulation-container">
717       <div class="track" id="simulation-track">
718         <div class="road-line"></div>
719         <div class="car-emoji" id="simulation-car">🚗</div>
720       </div>
721     </div>
722   </div>
723   <!-- Custom Resize Handles -->
724   <div class="resizer resizer-bl" id="resizer-bl"></div>
725   <div class="resizer resizer-br" id="resizer-br"></div>
726 </div>
727
728 </div>
729
730 <!-- LOGIKA JAVASCRIPT -->
731 <script>
732   // Fungsi formatter angka (Max 2 desimal, hapus nol tidak perlu)
733   function fmt(num) {
734     return Number(parseFloat(num).toFixed(2));
735   }
736
737   // =====
738   // FLOATING WIDGET DRAG & MINIMIZE LOGIC
739   // =====
740   function toggleSimWidget() {
741     const widget = document.getElementById('sim-widget');
742     const btn = document.getElementById('btn-minimize-sim');
743     widget.classList.toggle('minimized');
744
745     if (widget.classList.contains('minimized')) {
746       btn.innerText = '+';
747     } else {
748       btn.innerText = '-';
749     }
750   }
751
752   // Script to make widget draggable
753   dragElement(document.getElementById("sim-widget"));
754
755   function dragElement(elmnt) {
756     var pos1 = 0, pos2 = 0, pos3 = 0, pos4 = 0;
757     if (document.getElementById(elmnt.id + "-header")) {

```

```
758         document.getElementById(elmnt.id + "-header").onmousedown = dragMouseDown;
759     } else {
760         elmnt.onmousedown = dragMouseDown;
761     }
762
763     function dragMouseDown(e) {
764         e = e || window.event;
765         if (e.target.tagName.toLowerCase() === 'button' ||
e.target.classList.contains('resizer')) return;
766
767         e.preventDefault();
768
769         var rect = elmnt.getBoundingClientRect();
770         elmnt.style.top = rect.top + "px";
771         elmnt.style.left = rect.left + "px";
772         elmnt.style.bottom = "auto";
773         elmnt.style.right = "auto";
774
775         pos3 = e.clientX;
776         pos4 = e.clientY;
777         document.onmouseup = closeDragElement;
778         document.onmousemove = elementDrag;
779     }
780
781     function elementDrag(e) {
782         e = e || window.event;
783         e.preventDefault();
784         pos1 = pos3 - e.clientX;
785         pos2 = pos4 - e.clientY;
786         pos3 = e.clientX;
787         pos4 = e.clientY;
788
789         elmnt.style.top = (elmnt.offsetTop - pos2) + "px";
790         elmnt.style.left = (elmnt.offsetLeft - pos1) + "px";
791     }
792
793     function closeDragElement() {
794         document.onmouseup = null;
795         document.onmousemove = null;
796     }
797 }
798
799 // =====
800 // CUSTOM RESIZE LOGIC (KIRI & KANAN BAWAH)
801 // =====
802 function makeResizable(elmnt) {
803     const resizerBR = elmnt.querySelector('.resizer-br');
804     const resizerBL = elmnt.querySelector('.resizer-bl');
805
806     let original_width = 0, original_height = 0, original_x = 0, original_y = 0;
```

```
807     let original_mouse_x = 0, original_mouse_y = 0;
808
809     function lockPosition() {
810         var rect = elmnt.getBoundingClientRect();
811         elmnt.style.top = rect.top + "px";
812         elmnt.style.left = rect.left + "px";
813         elmnt.style.bottom = "auto";
814         elmnt.style.right = "auto";
815     }
816
817     resizerBR.addEventListener('mousedown', function(e) {
818         e.preventDefault();
819         if (elmnt.classList.contains('minimized')) return;
820         lockPosition();
821         original_width = parseFloat(getComputedStyle(elmnt,
null).getPropertyValue('width').replace('px', ''));
822         original_height = parseFloat(getComputedStyle(elmnt,
null).getPropertyValue('height').replace('px', ''));
823         original_mouse_x = e.pageX;
824         original_mouse_y = e.pageY;
825         window.addEventListener('mousemove', resizeBR);
826         window.addEventListener('mouseup', stopResize);
827     });
828
829     function resizeBR(e) {
830         const width = original_width + (e.pageX - original_mouse_x);
831         const height = original_height + (e.pageY - original_mouse_y);
832         if (width > 350) elmnt.style.width = width + 'px';
833         if (height > 150) elmnt.style.height = height + 'px';
834     }
835
836     resizerBL.addEventListener('mousedown', function(e) {
837         e.preventDefault();
838         if (elmnt.classList.contains('minimized')) return;
839         lockPosition();
840         original_width = parseFloat(getComputedStyle(elmnt,
null).getPropertyValue('width').replace('px', ''));
841         original_height = parseFloat(getComputedStyle(elmnt,
null).getPropertyValue('height').replace('px', ''));
842         original_x = elmnt.getBoundingClientRect().left;
843         original_mouse_x = e.pageX;
844         original_mouse_y = e.pageY;
845         window.addEventListener('mousemove', resizeBL);
846         window.addEventListener('mouseup', stopResize);
847     });
848
849     function resizeBL(e) {
850         const width = original_width - (e.pageX - original_mouse_x);
851         const height = original_height + (e.pageY - original_mouse_y);
852         if (width > 350) {
853             elmnt.style.width = width + 'px';
```

```
854         elmnt.style.left = original_x + (e.pageX - original_mouse_x) + 'px';
855     }
856     if (height > 150) elmnt.style.height = height + 'px';
857 }
858
859 function stopResize() {
860     window.removeEventListener('mousemove', resizeBR);
861     window.removeEventListener('mousemove', resizeBL);
862 }
863 }
864
865 makeResizable(document.getElementById("sim-widget"));
866
867 // =====
868 // UI Navigation Logic (Menu & Modals)
869 // =====
870 function showApp() {
871     document.getElementById('menu-container').style.display = 'none';
872     document.getElementById('app-container').style.display = 'block';
873     window.scrollTo(0, 0);
874 }
875
876 function showMenu() {
877     document.getElementById('app-container').style.display = 'none';
878     document.getElementById('menu-container').style.display = 'flex';
879 }
880
881 function showModal(modalId) {
882     document.getElementById(modalId).classList.add('active'); }
883
884 function closeModal(modalId) {
885     document.getElementById(modalId).classList.remove('active'); }
886
887 // =====
888 // App Calculation Logic
889 // =====
890 let currentMode = 'fisika';
891 let chartVelInstance = null;
892 let chartPosInstance = null;
893 let animId = null;
894
895 function setMode(mode) {
896     currentMode = mode;
897     const tabFisika = document.getElementById('tab-fisika');
898     const tabCustom = document.getElementById('tab-custom');
899     const formFisika = document.getElementById('form-fisika');
900     const formCustom = document.getElementById('form-custom');
901
902     const formulaPhysics = document.getElementById('physics-formula-ref');
903     const formulaCustom = document.getElementById('custom-formula-ref');
```

```

902     if (mode === 'fisika') {
903         tabFisika.classList.add('active');
904         tabCustom.classList.remove('active');
905         formFisika.style.display = 'block';
906         formCustom.style.display = 'none';
907         formulaPhysics.style.display = 'flex';
908         formulaCustom.style.display = 'none';
909     } else {
910         tabCustom.classList.add('active');
911         tabFisika.classList.remove('active');
912         formCustom.style.display = 'block';
913         formFisika.style.display = 'none';
914         formulaPhysics.style.display = 'none';
915         formulaCustom.style.display = 'flex';
916     }
917 }
918
919 function validateInputs() {
920     let t0, t1;
921     if (currentMode === 'fisika') {
922         const v0 = document.getElementById('input-v0').value;
923         const a = document.getElementById('input-a').value;
924         t0 = document.getElementById('input-t0').value;
925         t1 = document.getElementById('input-t1').value;
926         const s0 = document.getElementById('input-s0').value;
927         if (v0 === '' || a === '' || t0 === '' || t1 === '' || s0 === '') return
"SYSTEM ERROR: SEMUA FIELD HARUS DIISI!";
928     } else {
929         let rawFuncStr = document.getElementById('input-v-func').value;
930         const funcStr = rawFuncStr.replace(/(\d)([a-zA-Z\(\)]/g, '$1*$2').replace(/\)
([a-zA-Z0-9])/g, ')*$1');
931
932         t0 = document.getElementById('input-tc0').value;
933         t1 = document.getElementById('input-tc1').value;
934         const s0 = document.getElementById('input-sc0').value;
935         if (rawFuncStr === '' || t0 === '' || t1 === '' || s0 === '') return "SYSTEM
ERROR: SEMUA FIELD HARUS DIISI!";
936         try { nerdamer(funcStr); } catch (error) { return "SYNTAX ERROR: FUNGSI
MATEMATIKA TIDAK VALID!"; }
937     }
938     if (parseFloat(t1) <= parseFloat(t0)) return "LOGIC ERROR: WAKTU AKHIR (t1)
HARUS LEBIH BESAR!";
939     return null;
940 }
941
942 function generateLangkahPenyelesaian(params) {
943     let html = "";
944     if (currentMode === 'fisika') {
945         const { v0, a, t0, t1, s0, finalPos } = params;
946

```

```

947         html += `<strong>[STEP 1] Persamaan Umum Integral Terhadap Waktu:</strong>
<br>`;
948         html += `<i>s(t) = s0 + ∫ v(t) dt</i><br><br>`;
949
950         html += `<strong>[STEP 2] Memasukkan Fungsi Kecepatan Konstan (GLBB):
</strong><br>`;
951         html += `> Kecepatan awal (v0) = ${fmt(v0)}, Percepatan (a) = ${fmt(a)}
<br>`;
952         html += `<i>v(t) = ${fmt(v0)} + ${fmt(a)}t</i><br><br>`;
953
954         html += `<strong>[STEP 3] Melakukan Operasi Integral (Aturan Penambahan
Pangkat):</strong><br>`;
955         html += `> Aturan Integral Pangkat: <strong>∫ tn dt = [ 1 / (n+1) ]
tn+1</strong><br>`;
956         html += `> Memisahkan suku fungsi: ∫ (${fmt(v0)} + ${fmt(a)}t) dt = ∫
${fmt(v0)} t0 dt + ∫ ${fmt(a)} t1 dt<br>`;
957         html += `> Proses integral Suku ke-1 (${fmt(v0)}):<br>`;
958         html += `<i>= ${fmt(v0)} × [ 1 / (0+1) ] t0+1 = ${fmt(v0)}t</i><br>`;
959         html += `> Proses integral Suku ke-2 (${fmt(a)}t):<br>`;
960         html += `<i>= ${fmt(a)} × [ 1 / (1+1) ] t1+1 = ${fmt(a/2)}t2</i><br>`;
961         html += `> Hasil persamaan anti-turunan:<br>`;
962         html += `<i>S(t) = ${fmt(v0)}t + ${fmt(a/2)}t2</i><br><br>`;
963
964         html += `<strong>[STEP 4] Integral Tentu (Evaluasi Batas Waktu):</strong>
<br>`;
965         html += `> Menggunakan rumus evaluasi batas: s(t1) - s(t0)<br>`;
966         html += `> Menghitung nilai pada Batas Atas (t1 = ${fmt(t1)}):<br>`;
967         html += `<i>S(${fmt(t1)}) = ${fmt(v0)}(${fmt(t1)}) + ${fmt(a/2)}
(${fmt(t1)})2</i><br>`;
968         let st1_term1 = v0 * t1;
969         let st1_term2 = (a/2) * Math.pow(t1, 2);
970         let st1 = st1_term1 + st1_term2;
971         html += `<i>S(${fmt(t1)}) = ${fmt(st1_term1)} + ${fmt(st1_term2)} =
${fmt(st1)}</i><br>`;
972
973         html += `> Menghitung nilai pada Batas Bawah (t0 = ${fmt(t0)}):<br>`;
974         html += `<i>S(${fmt(t0)}) = ${fmt(v0)}(${fmt(t0)}) + ${fmt(a/2)}
(${fmt(t0)})2</i><br>`;
975         let st0_term1 = v0 * t0;
976         let st0_term2 = (a/2) * Math.pow(t0, 2);
977         let st0 = st0_term1 + st0_term2;
978         html += `<i>S(${fmt(t0)}) = ${fmt(st0_term1)} + ${fmt(st0_term2)} =
${fmt(st0)}</i><br>`;
979
980         let deltaS = st1 - st0;
981         html += `> Mengurangkan batas (S(t1) - S(t0)):<br>`;
982         html += `<i>Δs = ${fmt(st1)} - ${fmt(st0)} = ${fmt(deltaS)} m</i><br><br>`;
983
984         html += `<strong>[FINAL] Penjumlahan Posisi Awal dan Jarak:</strong><br>`;
985         html += `> Posisi Akhir = Posisi Awal + Perubahan Jarak<br>`;
986         html += `<i>s(${fmt(t1)}) = ${fmt(s0)} + ${fmt(deltaS)}</i><br>`;

```

```

987         html += `s(${fmt(t1)}) = ${fmt(s0 + deltaS)} m</i>`;
988
989     } else {
990         // MODIFIKASI: Menampilkan output penyelesaian Integral Tentu secara
eksplisit
991         const { funcStr, integralStr, t0, t1, s0, S_t0, S_t1, deltaS, finalPos } =
params;
992
993         html += `[STEP 1] Persamaan Jarak (Integral Tentu):</strong><br>`;
994         html += `s(t) =  $s_0 + \int_{t_0}^{t_1} v(t) dt$ </i><br>`;
995         html += `> Fungsi Kecepatan: v(t) = ${funcStr}</i><br><br>`;
996
997         html += `[STEP 2] Mencari Anti-Turunan S(t):</strong><br>`;
998         html += `S(t) =  $\int$  (${funcStr}) dt</i><br>`;
999         html += `S(t) = ${integralStr}</i><br><br>`;
1000
1001         html += `[STEP 3] Evaluasi Integral Tentu pada Batas Waktu:</strong>
<br>`;
1002         html += `>  $\Delta s = S(t_1) - S(t_0)$ <br>`;
1003         html += `> Evaluasi Batas Atas ( $t_1 =$ ${fmt(t1)})<br>`;
1004         html += `< S(${fmt(t1)}) = ${fmt(S_t1)}</i><br>`;
1005         html += `> Evaluasi Batas Bawah ( $t_0 =$ ${fmt(t0)})<br>`;
1006         html += `< S(${fmt(t0)}) = ${fmt(S_t0)}</i><br>`;
1007         html += `> Perubahan Jarak ( $\Delta s$ ):<br>`;
1008         html += `<  $\Delta s =$ ${fmt(S_t1)} - (${fmt(S_t0)})</i><br>`;
1009         html += `<  $\Delta s =$ ${fmt(deltaS)} m</i><br><br>`;
1010
1011         html += `[FINAL] Menghitung Posisi Akhir:</strong><br>`;
1012         html += `> s(${fmt(t1)}) =  $s_0 + \Delta s$ <br>`;
1013         html += `s(${fmt(t1)}) = ${fmt(s0)} + (${fmt(deltaS)})</i><br>`;
1014         html += `s(${fmt(t1)}) = ${fmt(finalPos)} m</i>`;
1015     }
1016     return html;
1017 }
1018
1019 function calculateData() {
1020     const nPoints = 100;
1021     let timeData = [], velData = [], posData = [];
1022     let totalJarak = 0, t0, t1, s0, dt, v0, a, funcStr;
1023     let params = {};
1024
1025     if (currentMode === 'fisika') {
1026         v0 = parseFloat(document.getElementById('input-v0').value);
1027         a = parseFloat(document.getElementById('input-a').value);
1028         t0 = parseFloat(document.getElementById('input-t0').value);
1029         t1 = parseFloat(document.getElementById('input-t1').value);
1030         s0 = parseFloat(document.getElementById('input-s0').value);
1031         dt = (t1 - t0) / nPoints;
1032         for (let i = 0; i <= nPoints; i++) {
1033             let t = t0 + i * dt;
1034             let v_t = v0 + a * (t - t0);

```

```

1035         let s_t = s0 + v0 * (t - t0) + 0.5 * a * Math.pow(t - t0, 2);
1036         timeData.push(fmt(t)); velData.push(v_t); posData.push(s_t);
1037         if (i > 0) totalJarak += Math.abs(posData[i] - posData[i-1]);
1038     }
1039     const finalPosition = posData[nPoints];
1040     params = { v0, a, t0, t1, s0, finalPos: finalPosition };
1041 } else {
1042     // MODIFIKASI: Kalkulasi dengan Integral Tentu (Definite Integral)
1043     let rawFuncStr = document.getElementById('input-v-func').value;
1044     funcStr = rawFuncStr.replace(/(\d)([a-zA-Z\(\)]/g, '$1*$2').replace(/\)([a-
zA-Z0-9])/g, ')*$1');
1045
1046     t0 = parseFloat(document.getElementById('input-tc0').value);
1047     t1 = parseFloat(document.getElementById('input-tc1').value);
1048     s0 = parseFloat(document.getElementById('input-sc0').value);
1049     dt = (t1 - t0) / nPoints;
1050
1051     // 1. Eksekusi Anti-Turunan menggunakan Nerdamer
1052     const integralExpr = nerdamer(`integrate(${funcStr}, t)`);
1053     const integralFunc = integralExpr.buildFunction(['t']);
1054
1055     // 2. Evaluasi pada batas bawah (t0) untuk proses perhitungan delta secara
    bertahap (untuk grafik)
1056     const S_t0 = integralFunc(t0);
1057
1058     timeData.push(fmt(t0));
1059
1060     let firstV = nerdamer(funcStr).evaluate({t: t0}).text();
1061     firstV = parseFloat(firstV);
1062     velData.push(firstV);
1063     posData.push(s0);
1064
1065     for (let i = 1; i <= nPoints; i++) {
1066         let t = t0 + i * dt;
1067
1068         // Evaluasi Kecepatan eksak
1069         let v_t = nerdamer(funcStr).evaluate({t: t}).text();
1070         v_t = parseFloat(v_t);
1071
1072         // Evaluasi Posisi eksak dari fungsi analitik dengan Integral Tentu s(t)
    = s0 + ( S(t) - S(t0) )
1073         let S_t = integralFunc(t);
1074         let s_t = s0 + (S_t - S_t0);
1075
1076         timeData.push(fmt(t));
1077         velData.push(v_t);
1078         posData.push(s_t);
1079
1080         totalJarak += Math.abs(posData[i] - posData[i-1]);
1081     }
1082     const finalPosition = posData[nPoints];

```

```

1083     const S_t1 = integralFunc(t1);
1084     const deltaS = S_t1 - S_t0;
1085
1086     params = { funcStr, integralStr: integralExpr.text(), t0, t1, s0, S_t0,
S_t1, deltaS, finalPos: finalPosition };
1087 }
1088     const finalPosition = posData[nPoints];
1089     return { timeData, velData, posData, totalJarak, finalPosition, params };
1090 }
1091
1092 function renderCharts(labels, dataV, dataS) {
1093     if (chartVelInstance) chartVelInstance.destroy();
1094     if (chartPosInstance) chartPosInstance.destroy();
1095
1096     Chart.defaults.color = '#000';
1097     Chart.defaults.font.family = 'Montserrat';
1098     Chart.defaults.font.weight = 'bold';
1099
1100     const commonOptions = {
1101         animation: false, responsive: true, maintainAspectRatio: false,
1102         plugins: { legend: { display: false } },
1103         scales: {
1104             x: { grid: { color: 'rgba(0,0,0,0.1)' } },
1105             y: { grid: { color: 'rgba(0,0,0,0.1)' } }
1106         }
1107     };
1108
1109     const ctxV = document.getElementById('chart-velocity').getContext('2d');
1110     chartVelInstance = new Chart(ctxV, {
1111         type: 'line',
1112         data: {
1113             labels: labels,
1114             datasets: [{
1115                 data: dataV, borderColor: '#00e5ff', backgroundColor: 'rgba(0, 229,
255, 0.2)',
1116                 borderWidth: 4, pointRadius: 0, fill: true, tension: 0.1, stepped:
false
1117             }]
1118         }, options: commonOptions
1119     });
1120
1121     const ctxS = document.getElementById('chart-position').getContext('2d');
1122     chartPosInstance = new Chart(ctxS, {
1123         type: 'line',
1124         data: {
1125             labels: labels,
1126             datasets: [{
1127                 data: dataS, borderColor: '#cc0044', backgroundColor: 'rgba(204, 0,
68, 0.2)',
1128                 borderWidth: 4, pointRadius: 0, fill: true, tension: 0.1
1129             }]

```

```
1130         }, options: commonOptions
1131     });
1132 }
1133
1134 function runSimulation(posData) {
1135     if (animId) cancelAnimationFrame(animId);
1136     const car = document.getElementById('simulation-car');
1137     const track = document.getElementById('simulation-track');
1138     const carWidth = 60;
1139     const minPos = Math.min(...posData);
1140     const maxPos = Math.max(...posData);
1141     let range = maxPos - minPos;
1142     if (range === 0) range = 1;
1143
1144     const duration = 2500;
1145     let startTime = null;
1146
1147     document.getElementById('sim-widget').style.display = 'flex';
1148
1149     function animate(timestamp) {
1150         if (!startTime) startTime = timestamp;
1151         let elapsed = timestamp - startTime;
1152         let progress = elapsed / duration;
1153
1154         if (progress > 1) progress = 1;
1155
1156         let exactIndex = progress * (posData.length - 1);
1157         let indexFloor = Math.floor(exactIndex);
1158         let indexCeil = Math.ceil(exactIndex);
1159         let indexFraction = exactIndex - indexFloor;
1160
1161         let currentPos = posData[indexFloor];
1162         if (indexCeil < posData.length) {
1163             currentPos += (posData[indexCeil] - posData[indexFloor]) *
indexFraction;
1164         }
1165
1166         let currentTrackWidth = track.clientWidth;
1167         let availableSpace = currentTrackWidth - carWidth - 40;
1168         if (availableSpace < 0) availableSpace = 0;
1169
1170         let normalizedScale = (currentPos - minPos) / range;
1171         let pixelTranslation = normalizedScale * availableSpace;
1172
1173         car.style.transform = `translateY(-50%) skewX(5deg)
translateX(${pixelTranslation}px) scaleX(-1)`;
1174
1175         if (progress < 1) {
1176             animId = requestAnimationFrame(animate);
1177         }
1178     }
1179 }
```

```
1178     }
1179     animId = requestAnimationFrame(animate);
1180 }
1181
1182 function mulaiSimulasi() {
1183     const errorMsgEl = document.getElementById('error-msg');
1184     const validationError = validateInputs();
1185     if (validationError) {
1186         errorMsgEl.innerText = validationError;
1187         errorMsgEl.style.display = 'block';
1188         return;
1189     }
1190     errorMsgEl.style.display = 'none';
1191
1192     const { timeData, velData, posData, totalJarak, finalPosition, params } =
calculateData();
1193     document.getElementById('output-jarak').innerText = fmt(totalJarak);
1194     document.getElementById('output-posisi').innerText = fmt(finalPosition);
1195     document.getElementById('langkah-penyelesaian').innerHTML =
generateLangkahPenyelesaian(params);
1196
1197     renderCharts(timeData, velData, posData);
1198     runSimulation(posData);
1199 }
1200 </script>
1201 </body>
1202 </html>
```